You Should Plug Your Telemed Ultrasound to Mains Power

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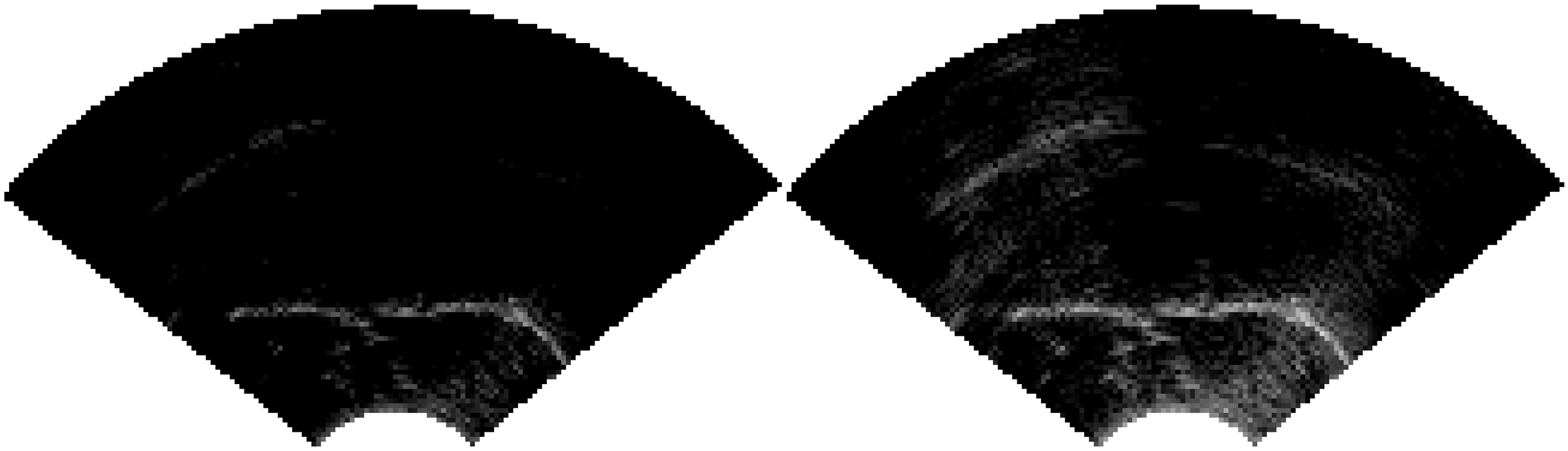
Introduction

- Ultrasound has been used for speech studies for decades (Shawker and Sonies 1985, Gick 2002, Stone 2005)
- In recent years Telemed micro ultrasound devices have become something of a norm.
- While the manual says to use mains power (connect the ultrasound device's power source to a wall socket) it does not say why, and the device will work with just power from the USB-connection to a computer.
- We discovered by accident why this recommendation exists: When running on USB power the ultrasound device will raise imaging power by steps during each recording.
- According to Telemed, this is done to avoid setting off a short circuit panic-mode in the USB port.
- We report technical testing results obtained by using a bovine steak as a phantom.

Methods: Recording

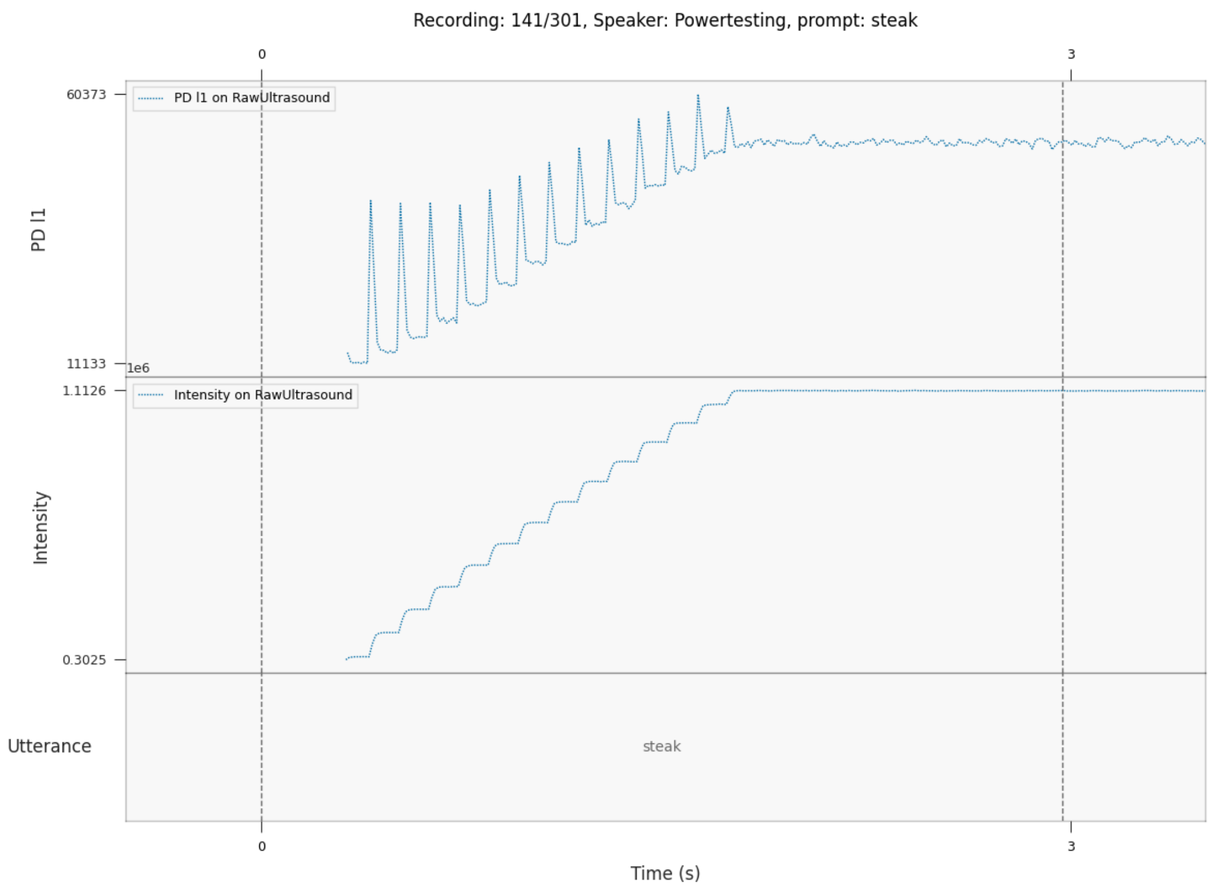
- The conditions in the recordings were
  - **without** mains power
    - only air in contact with the probe – 100 recordings
    - stationary phantom – 100 recordings
    - phantom moved manually by experimenter – 23 recordings
  - **with** mains power
    - stationary phantom – 13 recordings
    - phantom moved manually by experimenter – 25 recordings
- Data was analysed in PATKIT (Palo et al. 2026).

Results: Image brightness



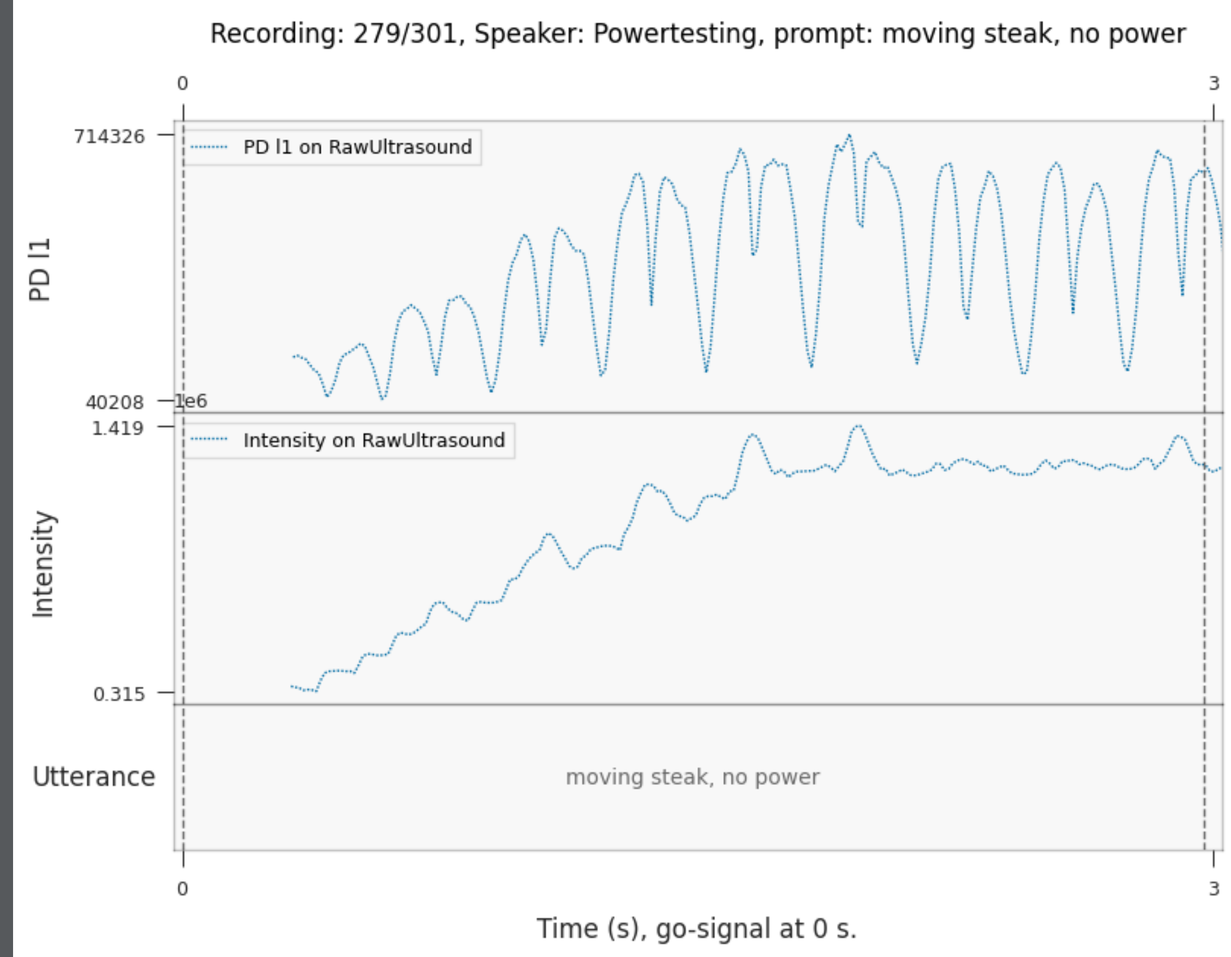
- Ultrasound frames at the beginning of the recording (left) and at the end of the intensity ramp (right).
- While neither the probe nor the phantom have moved, the image on the right is considerably brighter.

Results: Pixel Difference (PD) artefacts in stationary phantom



- Panels from top to bottom: Pixel Difference (PD), Image overall ultrasound intensity, Sound spectrogram, Sound waveform, and TextGrid (showing only the prompt 'hold still').
- As can be seen the image intensity ramps up by steps from beginning of the ultrasound recording at about $t = 0.3$ s to about $t = 1.7$ s, at which point the intensity plateaus.
- There is a series of spikes in the PD curve which correspond to the moments when the intensity rises a step.

Results: Pixel Difference (PD) of moving phantom



- The intensity pattern remains.
- Spikes in the PD curve are now hidden but the over all amplitude grows as intensity grows.

Discussion: Data quality

- Unfortunately, the stepping of intensity has consequences for data quality.
  - Imaging with lower power produces darker images.
  - In practice this means that at the beginning of the recording the ultrasound image will brighten in steps.
  - This may or may not affect anatomical tracking, and can cause artefacts in analysis metrics such as Pixel Difference (PD) (Palo 2019) and other analysis methods.

References

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Shawker, T. H. and Sonies, B. C. (1985). Ultrasound biofeedback for speech training. Instrumentation and preliminary results. *Investigative Radiology*, 20(1):90–93.

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Conclusions

- For new data: Plug the ultrasound machine into mains power whenever possible and if you need to run it with only USB power, make a note of that in your recording metadata and/or laboratory journal.
- For old data, if you do not know that the device was plugged in to mains electricity, there are two ways of checking if there is a problem with data quality.
- First, play a recording and look at the data. If the ultrasound image seems to get brighter they may be a problem.
- Second, examine the intensity curves in PATKIT. Depending on image contents, you may need to look at more than one recording to see if intensity is going up step-wise.
- In case the intensity is ramping up, any intensity based measures might be unreliable.
- Examining only PD curves in PATKIT or AAA (Articulate Instruments 2012) is not reliable as seen in the moving phantom case.
- Not seeing the pattern in the PD curve of a recording should not be taken as evidence of the absence of the problem in a dataset, nor even the recording in question, as it may affect event timing without being easily detectable.
- So, plug your ultrasound to mains power.

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